

CHAPTER - II

DATA ABSTRACTION

2.1 :::: Data Abstraction - Introduction:

- i). Data Abstraction is powerful concept in Computer Science
- ii). Data Abstraction allows programmers to treat code as object. For example car object, pencil object etc.
- iii). Programmers need not worry about how code is implemented.
- iv). Programmers just know what object does.
- v). **Abstraction** provides modularity
- vi). **Modularity** means splitting a program into many modules



- vii). **Classes (Structures)** are the representation for "Abstract Data Types (ADT)"

2.2 :::: Abstract Data Type:

2m* / 3m

- i). ADT stands for Abstract Data Type
- ii). ADT is a type or class for objects whose behavior is defined by a set of value or a set of operations.
- iii). ADT defines what operations are to be performed.
- iv). ADT not defines how operations will be performed.
- v). ADT not specifies how data will be organized in the memory.
- vi). List ADT can be implemented by using
 - (a) Single Linked List
 - (b) Doubly Linked List
 - (c) Stack ADT
 - (d) Queues ADT

Abstraction:

- i). The process of providing only the essentials and hiding the details is known as abstraction.
- ii). Abstraction replicates how we think about the world. For example, for drive a car, no need to know about engine or tire details etc.
- iii). Abstraction gives implementation independent view.

2.3 :::: Constructor and Selector:

- Data abstraction is achieved by defining Abstract data type.
- Abstract data type is a collection of constructors and selectors.
- There are two types of functions which facilitate data abstraction. They are,

a. Constructor

b. Selector

*Differentiate***Constructor:** & *selector :- 2 m &*

- i). Constructors are functions which build abstract data type.
- ii). Constructors are used to create an object which combines different pieces of information.
- iii). Example:

city=makecity(name,lat,lon)

(name, lat, lon) → value passed as parameter

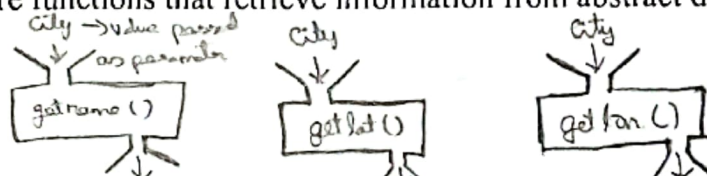


In the above example,

- Makecity()** is a constructor which is used to create an object named city.
- City object** combines three informations such as
 - o **Name** of the city
 - o **Longitude** of the city
 - o **Latitude** of the city

Selector:

- i). Selectors are functions that retrieve information from abstract data type.
- ii). Example:



- `getname(city)` extract name of the city from city object.
- `getlat(city)` extract latitude of the city from city object.
- `getlon(city)` extract longitude of the city from city object.

2.4 :::: Representation of Abstract Data Type using Rational Numbers:

i). The basic idea of data abstraction is to structure programs operate on abstract data type.

ii). Concrete data representation is defined as an independent part of the program.

iii). In concrete data representation, a definition for each function is known.

iv). Any program which consists of two parts. They are,

(1) The part operates on Abstract Data

(2) The part defines concrete representation

a). Concrete representation is connected by small set of functions which implement Abstract Data.

vi). **Example:**

(a) A rational number is a ratio of integers.

(b) Rational numbers are important sub class of real numbers.

(c) The rational number $8/3$ is written as Numerator/Denominator

(d) Both parts Numerator and Denominator requires to characterize the value of rational number.

(e) Dividing integers produce float result which losing exact precision of integers

(f) $8/3$ returns 2.66666

✓Wishful Thinking:

i). Wishful thinking is a powerful strategy for designing programs.

ii). Wishful thinking is the formation of beliefs.

iii). Wishful thinking making a decision according to what might be pleasing to imagine instead of appealing to reality.

2.5 :: Lists & Tuples:

- i). Python provides compound structure called pair.
- ii). Pair is made up of list or tuple. *2m*

Bundling two values together into one can be considered as a pair.

List: *2m*

- i). List is constructed by placing expressions within square bracket separated by commas.
- ii). An expression in list is called as List Literal.
- iii). List can store multiple values.
- iv). Each value in list can be any data type.
- v). User can change the elements of the list.
- vi). Example:

`list1=[10,20]`

- oii). The elements of the list in two ways. They are

- (a) Multiple Assignments
- (b) Element selection operator

(a) Multiple Assignment:

`list1 = [10,20]`

`x,y = list1`

Now, x has the value 10 and y has the value 20.

(b) Element selection operator:

`list = [10,20]`

`list[0]` gives the value 10.

`list[1]` gives the value 20

Mathematical Representation of List:

- Mathematically list can be represented like a set.

- Example:

list[(0,10), (1,20)]

In the above example,

- 0 and 1 represents index position
- 10 and 20 represent value
- The way of bundling two values together into one is called as **Pair**.
- List is a common method to do the pair. Therefore list is called as **Pair**.

Representation Rational Numbers Using List:

- User can represent rational number as a pair of two integers.

```
rational(n, d):  
    return [n, d]  
numer(x):  
    return x[0]  
denom(x):  
    return x[1]
```

Tuple: *em*

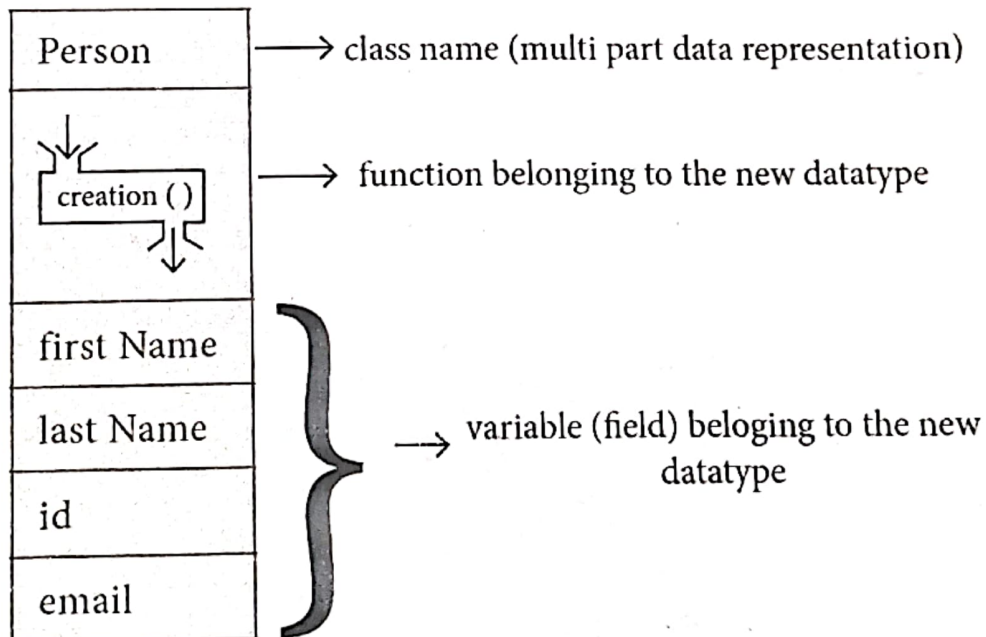
- i). Tuple is constructed by placing expressions within parenthesis separated by commas.
- ii). Tuple is similar to List.
- iii). Tuple can store multiple values.
- iv). The difference between tuple and list is, the elements of the tuple cannot be changed.
In List, the elements of the list can be changed if required.
- v). Example:

```
t=(10,20,30)  
t[0] gives the value 10.  
t[1] gives the value 20  
t[2] gives the value 30
```

2.6 :: Data Abstraction in Structure:

```

class Person:
    creation()
    firstName := ""
    lastName := ""
    id := ""
    email := ""
  
```



Person = ['Patra']

Let main() contain

p1:=Person()

firstName := "Pa

lastName := "Bas

id := "994-222-1

email="compsc

In the above table

- Person is referred
- P1 is referred
- Creation() is
- Firstname, last

Class:

A class

Point

- Abstract
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Person = ['Padmashri', 'Baskar', '994-222-1234', 'compsci@gmail.com']

Let main() contains

p1:=Person()	statement creates the object.
firstName := " Padmashri "	setting a field called firstName with value Padmashri
lastName := "Baskar"	setting a field called lastName with value Baskar
id := "994-222-1234"	setting a field called id value 994-222-1234
email="compsci@gmail.com"	setting a field called email with value compsci@gmail.com
- - output of firstName : Padmashri	

In the above table,

- Person is referred as a class.
- P1 is referred as an object.
- Creation() is a constructor
- Firstname, lastname, id and email are data (variable or field)

Class:

A class is a bundled data and function which work on that data.

Points to remember:

- Abstract Data type (ADT) is a type (or class) for objects whose behavior is defined by a set of value and a set of operations.
- The definition of ADT only mentions what operations are to be performed but not how these operations will be implemented.
- ADT does not specify how data will be organized in memory and what algorithms will be used for implementing the operations
- Constructors are functions that build the abstract data type.
- Selectors are functions that retrieve information from the data type.
- Concrete data types or structures (CDT's) are direct implementations of a relatively simple concept.
- Abstract Data Types (ADT's) offer a high level view (and use) of a concept independent of its implementation.

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Concrete data type

* In concrete data type representation is known

* It is defined as an independent part of the program

Abstract data type

* In Abstract data type representation is unknown.

* It is defined as a set of values or a set of operations.

Points to remember:

- A concrete data type is a data type whose representation is known and in abstract data type the representation of a data type is unknown
- Pair is a compound structure which is made up of list or Tuple
- List is constructed by placing expressions within square brackets separated by commas
- The elements of a list can be accessed in two ways. The first way is via multiple assignment and the second method is by the element selection operator
- Bundling two values together into one can be considered as a pair
- List does not allow to name the various parts of a multi-item object.



Evaluation

Part - I

Choose the best answer

(1 Mark)

1. Which of the following functions that build the abstract data type ?
(A) Constructors (B) Destructors (C) recursive (D) Nested
2. Which of the following functions that retrieve information from the data type?
(A) Constructors (B) Selectors (C) recursive (D) Nested
3. The data structure which is a mutable ordered sequence of elements is called
(A) Built in (B) List (C) Tuple (D) Derived data
4. A sequence of immutable objects is called
(A) Built in (B) List (C) Tuple (D) Derived data
5. The data type whose representation is known are called
(A) Built-in datatype (B) Derived datatype
(C) Concrete datatype (D) Abstract datatype
6. The data type whose representation is unknown are called
(A) Built in datatype (B) Derived datatype
(C) Concrete datatype (D) Abstract datatype
7. Which of the following is a compound structure?
(A) Pair (B) Triplet (C) single (D) quadrat

8. Bundling two values together into one can be considered as

- (A) Pair (B) Triplet (C) single (D) quadrat

9. Which of the following allow to name the various parts of a multi-item object?

- (A) Tuples (B) Lists (C) Classes (D) quadrats

10. Which of the following is constructed by placing expressions within square brackets?

- (A) Tuples (B) Lists (C) Classes (D) quadrats

Part - II

Answer the following questions

(2 Marks)

1. What is abstract data type?
2. Differentiate constructors and selectors.
3. What is a Pair? Give an example.
4. What is a List? Give an example.
5. What is a Tuple? Give an example.

Part - III

Answer the following questions

(3 Marks)

1. Differentiate Concrete data type and abstract datatype.
2. Which strategy is used for program designing? Define that Strategy.
3. Identify Which of the following are constructors and selectors?

- (a) N1:=number() (b) accetnum(n1) (c) displaynum(n1)
(d) eval(a/b) (e) x,y:= makeslope (m), makeslope(n)
(f) display()

a, e - constructors
b, c, d, f - selectors

4. What are the different ways to access the elements of a list. Give example.
5. Identify Which of the following are List, Tuple and class ?

- (a) arr [1, 2, 34] (b) arr (1, 2, 34) (c) student [rno, name, mark]
(d) day:= ('sun', 'mon', 'tue', 'wed') (e) x:= [2, 5, 6.5, [5, 6], 8.2]
(f) employee [eno, ename, esal, eaddress]

a, e - list
c, f - class
b, d - tuple

Part - IV

(5Marks)

Answer the following questions

1. How will you facilitate data abstraction. Explain it with suitable example
2. What is a List? Why List can be called as Pairs. Explain with suitable example
3. How will you access the multi-item. Explain with example.

Reference Books

1. *Data structure and algorithmic thinking with python by narasimha karumanchi*
2. *sign and analysis of algorithms by s sridhar*
3. *Data Structures and Algorithms in Python by Goodrich, Tamassia & Goldwasser*
4. <https://www.tutorialspoint.com>

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